

Evaluating Predictive Value of G-Cross in Head and Neck Squamous Cell Carcinoma

Instructor: Dr. Jean Morrison, UMID: 97480851

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Abstract

Introduction Established predictors of survival for Head and Neck Squamous Cell Carcinoma (HNSCC) include cancer stage, tumor site, HPV status, and tobacco and alcohol exposure. While the number of immune cells alone has shown inconsistent prognostic value in HNSCC, the distance between tumor cells and immune cells, called G-Cross, has been shown to be associated with overall survival. In this study, we investigated if G-Cross is significantly associated with survival and assessed its predictive value after adjusting for known predictors.

Statistical Methods The data came from two sources, The Cancer Genome Atlas (TCGA), and the University of Michigan (UMich). Missing data was imputed with Multiple Imputation using Chained Equations. Multivariable Cox proportional hazards models were used to model the hazard of death. TCGA data was used as training data for model selection, and reduced models were created with and without G-Cross included. We ran the reduced models on the UMich testing data, and compared C-index and Brier scores to evaluate the prediction value of G-Cross.

Results Higher G-Cross values were associated with improved survival after adjusting for age, tumor site, cancer stage, and extranodal extension. (Hazard Ratio: 0.88, 95% CI: 0.82, 0.95). In the UMich testing dataset, inclusion of G-Cross improved predictive performance, increasing the C-index from 0.646 to 0.673 and decreasing the five-year Brier score.

Conclusions Our study indicates that G-Cross was significantly associated with survival after adjustment for known predictors. Inclusion of G-Cross improved predictive performance in an external testing dataset, suggesting it may provide additional prognostic information for patients with HNSCC.

Introduction

Head & Neck Squamous Cell Carcinoma (HNSCC) is characterized by malignancies in the squamous epithelium lining of the oral cavity, tonsils, base of the tongue, hypopharynx, larynx, and sinonasal cavity. It is the seventh most common cancer globally. Known predictors of survival for HNSCC patients include cancer stage, tumor site, HPV status, and cigarette and alcohol use.

In other cancers, a higher number of Tumor-Infiltrating Lymphocytes (TILs) in the tumor microenvironment is associated with better prognosis. But in HNSCC, just the number of TIL's has been an inconsistent predictor. Instead, the distance between tumor cells and TILs has been shown to be a stronger predictor of overall survival. This distance between tumor cells and TILs is measured using the G-Cross score, with higher scores corresponding to closer proximity between a tumor and immune cell. Previous research has shown that higher G-Cross scores are significantly associated with increased survival, without adjusting for other covariates. In this study, we aimed to evaluate if G-Cross is significantly associated with survival and assess its predictive value, after adjusting for known predictors.

Statistical Methods

The data in this study comes from two sources. First, N=404 observations come from The Cancer Genome Atlas (TCGA), which includes participants from around the world. Second, N=265 observations come from The University of Michigan (UMich), which includes participants only in Michigan. Each participant has an overall survival time in days, a censoring indicator that specifies if the participant was observed to die or was censored, the G-Cross score, and other covariates.

Shown in Appendix Table A1 is the number and proportion of missing values of each variable. To address the missing data, we used Multiple Imputation using Chained Equa-

tions (MICE). The two datasets were imputed independently in order to avoid data leakage between them. Ten imputed datasets were generated for each dataset using predictive mean matching for continuous variables, logistic regression for binary variables, multinomial logistic regression for categorical variables, and proportional odds regression for ordinal variables.

We used the TCGA dataset as training data for model selection, because it includes a larger and more diverse patient population, allowing more stable estimation of model parameters. We used the UMich dataset as testing data for model evaluation, allowing for external validation of the model and assessment of generalizability to a separate cohort.

In order to evaluate if G-Cross is a significant predictor of survival after adjusting for known predictors, we used multivariable Cox proportional hazards models, which allow us to account for censoring and model the hazard of death. Each model was fitted on each of the 10 imputed datasets, and results were pooled according to Rubin’s rules. We tested the proportional hazards assumption using the Schoenfeld residual test, and there were no violations in either dataset.

The known predictors of survival included in the models, with reference groups in bold, are: gender (male or **female**), age, stage (**1**, 2, 3, or 4), HPV status (yes or **no**), cigarette history (yes or **no**), alcohol history (yes or **no**), tumor site (**larynx**, pharynx, oral cavity, or other), lymphovascular invasion (yes or **no**), and extranodal extension (yes or **no**). Lymphovascular invasion is an indicator for when tumor cells have entered nearby blood vessels. Extranodal extension is an indicator for when a tumor that starts in a lymph node spreads beyond the lymph node into surrounding tissue. G-Cross scores ranged from 0 to about 1.5. In order to improve interpretability, G-Cross was scaled by 0.1 for modeling. A high G-Cross cutoff was created at 0.25, but this was only used for visualization purposes; G-Cross was modeled as a continuous variable.

We first fit the full model with G-Cross and all other covariates in the TCGA training

data. Variables with p-values < 0.10 were retained in the reduced model. We used the threshold of 0.10 to avoid excluding potentially important predictors.

To evaluate the reduced model, we computed the C-index and Brier score. C-index measures the probability that the survival times of two individuals are correctly ordered by a predictive score. Larger values indicate better predictive discrimination. The C-index was calculated separately for each time the model was fit to the 10 imputed datasets, and the average C-index was reported. Brier score measures the accuracy of predicted probabilities at a specific follow-up time, and a smaller Brier score indicates a better fitting model. Brier scores for each of the ten iterations over all timepoints were calculated, and to summarize, average Brier score at 5 years was reported.

In order to evaluate the predictive value of G-Cross, we compared the C-index and Brier score from the model with G-Cross and significant covariates to the model without G-cross. Evaluating model performance only in the training data may lead to overly optimistic estimates because the model parameters are estimated from that same dataset. Therefore, we ran the model with G-Cross and significant covariates and the model without G-cross on the UMich testing data. Using the predictors from the TCGA training data, we calculated fitted probabilities for the UMich observations. We then compared the C-index and Brier score again to determine the predictive value of including G-Cross in the model and to evaluate the model’s generalizability.

All analyses were performed using R statistical software, version 4.5.2 (www.R-project.org). MICE was implemented using the mice package, and Cox proportional hazards models were implemented using the survival package.

Results

Summary statistics of the known predictors of survival by data source are shown in Table 1. There are differences in frequencies of variables between data sources, for example TCGA

has a higher proportion of participants with Stage 4 HNSCC, and a lower proportion of HPV positive participants. Kaplan-Meier curves by high and low G-Cross for each dataset are shown in Appendix Figure A1. Up to the 10 year point, we can see that those with high G-Cross scores generally have higher survival probabilities.

After creating the full model in the TCGA training dataset, the following variables had p-values < 0.1 and were included in the reduced model: G-Cross, age, tumor site, cancer stage, and extranodal extension. We estimate that for every 0.1 increase in G-Cross, the hazard of death is about 12% lower, adjusting for other variables (HR: 0.88, 95% CI: 0.82, 0.95). We can conclude that G-Cross is significantly associated with survival, after adjusting for known predictors. Fitted survival probabilities comparing Stage 1 to Stage 4 and high to low G-Cross scores is shown in Figure 1. The reduced model predicts that Stage 1 patients have overall higher predicted survival probabilities compared to Stage 4. And within the stages, people with high G-Cross have higher predicted survival probabilities compared to those with low G-Cross. Further results from the reduced model in the TCGA training set are shown in Table 2.

C-Index and Brier scores in both the training and testing sets are shown in Table 3. In the UMich testing set, the C-index with G-Cross included in the model was 0.673, and the mean Brier score at 5 years was 0.194. In the model without G-Cross, C-index was 0.646, and Brier score was 0.229. Based on the results in the testing set, we can conclude that G-Cross does add predictive value; including it in the model increased C-index and decreased Brier score in the testing set.

Discussion

This analysis examined if G-Cross is significantly associated with survival adjusting for other covariates, and if it improves prediction of survival among patients with HNSCC. After adjusting for known predictors of survival, including age, tumor site, cancer stage, and

extranodal extension, G-Cross was significantly associated with survival. Specifically, higher values of G-Cross were associated with an increased hazard of death. These findings suggest that G-Cross is a predictor of survival in HNSCC, adjusting for other covariates.

We also examined whether including G-Cross improved prediction. In the testing dataset, the model including G-Cross had a greater C-index and a lower Brier score at five years compared with the model without G-Cross. This suggests that incorporating G-Cross may improve model discrimination and overall prediction accuracy in an independent dataset. Overall, the magnitude of the Brier scores suggests moderate prediction error, with similar levels of model performance across the training and testing datasets. The discrimination of the models was modest, with C-index values ranging from approximately 0.65 to 0.68 across datasets. The modest C-index suggests that additional unmeasured predictors may be needed to substantially improve risk stratification.

One limitation of this study should be noted. The TCGA and UMich datasets were used separately for model training and testing rather than being combined and randomly split. While combining the datasets could have increased the sample size available for model development, maintaining separate datasets allowed evaluation of model performance in an independent cohort. This approach provides a stronger assessment of generalizability but may reduce statistical power for model estimation.

In conclusion, our study provides evidence that G-Cross is significantly associated with survival after adjusting for known predictors, and improves prediction in HNSCC patients. These findings suggest that G-Cross may provide additional prognostic information beyond established predictors.

Generative AI Statement

ChatGPT 5.2 was used in this report. Specifically, I used ChatGPT to help me with clarity in writing, code debugging, and formatting tables and figures. I did not use ChatGPT or any other AI tools for any other assistance.

Tables and Figures

	The Cancer Genome Atlas	University of Michigan
n	404	265
Age (Mean (SD))	62.15 (11.83)	66.90 (11.40)
Gender		
Female	111 (27.5)	75 (28.3)
Male	293 (72.5)	190 (71.7)
Cancer Stage		
Stage 1	20 (5.5)	11 (4.3)
Stage 2	59 (16.2)	89 (34.8)
Stage 3	64 (17.6)	42 (16.4)
Stage 4	221 (60.7)	114 (44.5)
HPV		
No	367 (90.8)	162 (68.1)
Yes	37 (9.2)	76 (31.9)
Cigarette History		
No	179 (44.3)	99 (37.5)
Yes	225 (55.7)	165 (62.5)
Alcohol History		
No	136 (34.5)	148 (56.1)
Yes	258 (65.5)	116 (43.9)
Tumor Site		
Larynx	94 (23.3)	21 (8.1)
Pharynx	44 (10.9)	64 (24.2)
Oral Cavity	213 (52.7)	148 (56.9)
Other	53 (13.1)	28 (10.8)
Lymphovascular invasion		
No	188 (65.5)	99 (34.5)
Yes	96 (36.6)	166 (63.4)
Extranodal extension		
No	208 (67.5)	219 (85.5)
Yes	100 (32.5)	37 (14.5)
G-Cross Score (Mean (SD))	0.26 (0.21)	0.18 (0.31)

Table 1: Characteristics of Participants Diagnosed with Head & Neck Squamous Cell Carcinoma by Data Source

Numeric summaries are “Count (Percentage)” for categorical variables and “Mean(SD)” for continuous variables.

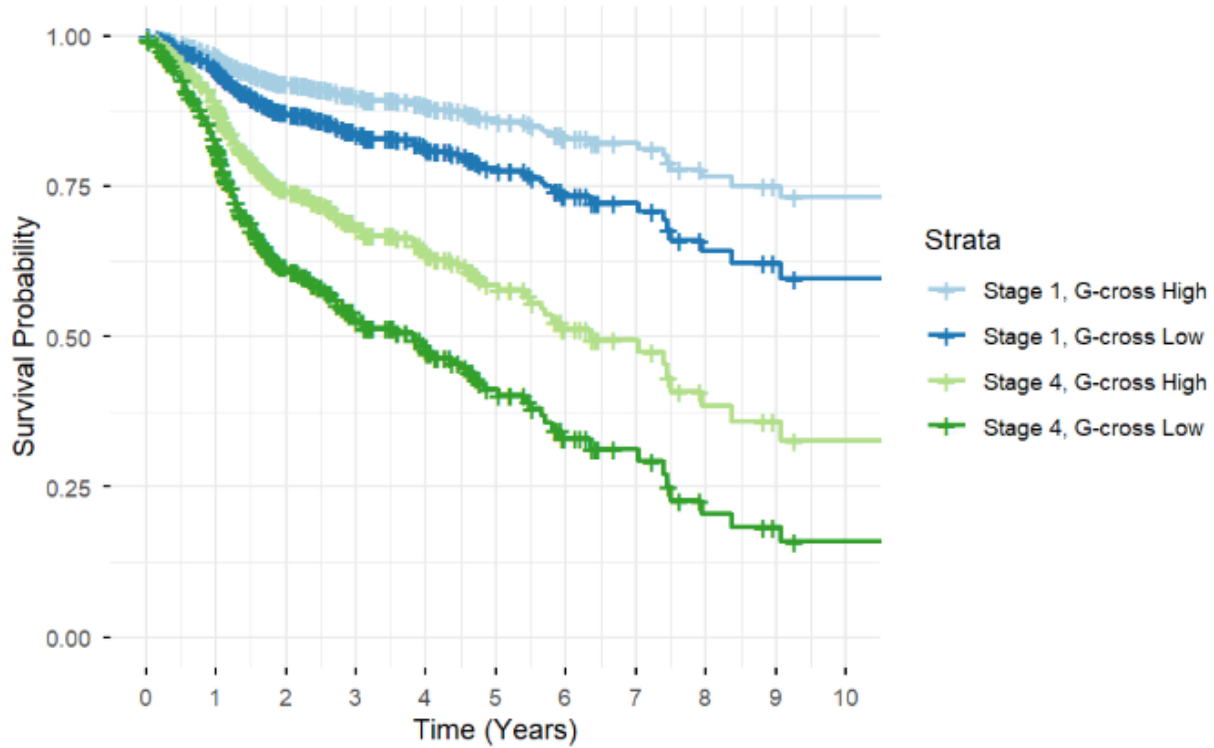


Figure 1: Fitted Survival Probabilities By Stage 1 vs 4 and High vs. Low G-Cross in TCGA Cohort

Note: High G-Cross is > 0.25 , low G-Cross is ≤ 0.25

Variable	Hazard Ratio	95% Confidence Interval
G-Cross (per 0.1 increase)	0.88	(0.82, 0.95)
Age	1.03	(1.02, 1.05)
Tumor Site (vs. Larynx)		
Oral Cavity	1.25	(0.85, 1.84)
Pharynx	1.21	(0.68, 2.15)
Other	1.48	(0.89, 2.46)
Cancer Stage (vs. Stage 1)		
2	1.92	(0.53, 6.95)
3	2.61	(0.73, 9.30)
4	2.97	(0.83, 10.60)
Extranodal Extension		
Yes (vs. No)	2.66	(1.79, 3.95)

Table 2: Results of Reduced Model from TCGA Training Data

Hazard ratios represent the multiplicative change in the hazard of death, adjusting for all other variables. Values > 1 indicate increased hazard, and values < 1 indicate decreased hazard.

Model	Mean C-Index	Mean Brier Score at 5 years
TCGA		
G-cross and significant covariates	0.683	0.214
Only significant covariates	0.669	0.225
UMich		
G-cross and significant covariates	0.673	0.194
Only significant covariates	0.646	0.229

Table 3: Prediction Measures of Models

Note: Significant covariates are those shown in Table 2: G-Cross, age, tumor site, cancer stage, and extranodal extension

Appendix

Variable	TCGA	UMich
Survival Time	0	0
G-Cross Score	0	0
Age	0	0
Gender	0	0
Cancer Stage	40 (9.9)	9 (3.4)
HPV	0	27 (10.2)
Cigarette History	0	1 (0.4)
Alcohol History	10 (2.5)	1 (0.4)
Tumor Site	0	5 (1.9)
Lymphovascular invasion	117 (29.0)	3 (1.1)
Extranodal extension	96 (23.8)	9 (3.4)

Table A1: Missingness Across Variables by Data Source

Each numeric summary is “Count (Percentage)” of missing observations.

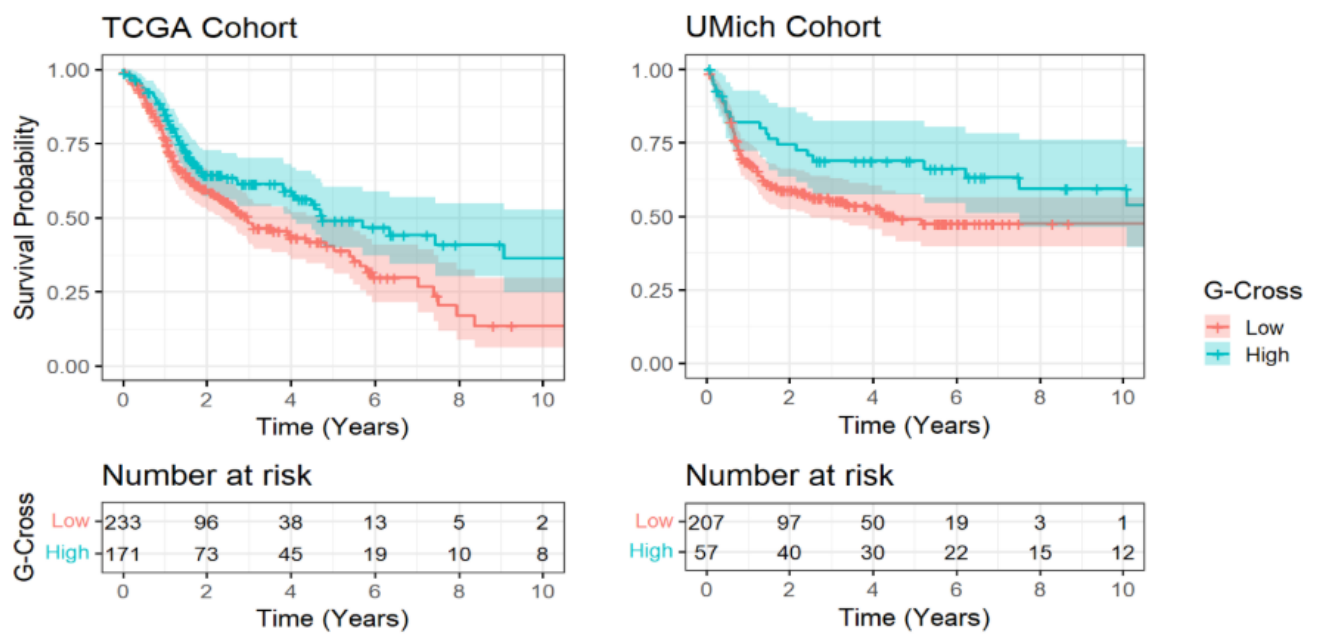


Figure A1: Kaplan-Meier Curves of Observed Survival Probabilities over Time by High vs. Low G-Cross in Each Cohort

Note: High G-Cross is > 0.25 , low G-Cross is ≤ 0.25